```
import torch
print(torch.cuda.is_available())
```

```
True
```

```
!pip install torch torchvision pandas matplotlib scikit-learn tqdm
```

```
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cu128)
Requirement already satisfied: torchvision in /usr/local/lib/python3.12/dist-packages (0.25.0+cu128)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (4.67.3)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: cuda-bindings==12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch) (12.9.4)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.93)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.90)
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Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from torchvision) (2.0.2)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from torchvision) (11.3.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
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Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
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Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.0)
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Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
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Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import pandas as pd

df = pd.read_csv("published_images.csv")

print(df.columns)
df.head()
```

```
Index(['uuid', 'iiifurl', 'iiifthumburl', 'viewtype', 'sequence', 'width',
       'height', 'maxpixels', 'openaccess', 'created', 'modified',
       'depictstmsobjectid', 'assistivetext'],
      dtype='object')
```

	uuid	iiifurl	iiifthumburl	viewtype	sequence	width	height	maxpixels	openaccess	cre
0	00007f61-4922-417b-8f27-893ea328206c	https://api.nga.gov/iiif/00007f61-4922-417b-8f...	https://api.nga.gov/iiif/00007f61-4922-417b-8f...	primary	0	3365	4332	NaN	1	201 15:4
1	0000bd8c-39de-4453-b55d-5e28a9beed38	https://api.nga.gov/iiif/0000bd8c-39de-4453-b5...	https://api.nga.gov/iiif/0000bd8c-39de-4453-b5...	primary	0	3500	4688	NaN	1	201 14:3
2	0001668a-dd1c-48e8-9267-b6d1697d43c8	https://api.nga.gov/iiif/0001668a-dd1c-48e8-92...	https://api.nga.gov/iiif/0001668a-dd1c-48e8-92...	primary	0	3446	4448	NaN	1	201 14:5
3	0002e210-8bc2-4471-bb0b-e7bb51d01a40	https://api.nga.gov/iiif/0002e210-8bc2-4471-bb...	https://api.nga.gov/iiif/0002e210-8bc2-4471-bb...	primary	0	2000	1571	900.0	0	200 17:2
4	00032658-8a7a-44e3-8bb8-df8c172f521d	https://api.nga.gov/iiif/00032658-8a7a-44e3-8b...	https://api.nga.gov/iiif/00032658-8a7a-44e3-8b...	primary	0	2674	3798	NaN	1	201 15:3

```
import torchvision.datasets as datasets
import torchvision.transforms as transforms

transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor()
])

dataset = datasets.CIFAR10(root='./data', download=True, transform=transform)
```

```
100%|██████████| 200/200 [04:28<00:00,  1.34s/it]
```

```
import torch
import torchvision.models as models

model = models.resnet18(pretrained=True)
model = torch.nn.Sequential(*list(model.children())[:-1])
model.eval()

def get_embedding(img):
    with torch.no_grad():
        emb = model(img.unsqueeze(0))
    return emb.flatten().numpy()
```

```
/usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py:208: UserWarning: The parameter 'pretrained' is deprecated since 0
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py:223: UserWarning: Arguments other than a weight enum or `None` for
  warnings.warn(msg)
```

```
import torchvision.datasets as datasets
import torchvision.transforms as transforms

transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor()
])

dataset = datasets.CIFAR10(root='./data', download=True, transform=transform)
```

```
100%|██████████| 170M/170M [00:49<00:00, 3.47MB/s]
```

```
import torch
import torchvision.models as models

model = models.resnet18(pretrained=True)
model = torch.nn.Sequential(*list(model.children())[:-1])
model.eval()
```

```
    (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
```

```
      (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (downsample): Sequential(
        (0): Conv2d(64, 128, kernel_size=(1, 1), stride=(2, 2), bias=False)
        (1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      )
    )
    (1): BasicBlock(
      (conv1): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (relu): ReLU(inplace=True)
      (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    )
  )
  (6): Sequential(
    (0): BasicBlock(
      (conv1): Conv2d(128, 256, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), bias=False)
      (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (relu): ReLU(inplace=True)
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (downsample): Sequential(
        (0): Conv2d(128, 256, kernel_size=(1, 1), stride=(2, 2), bias=False)
        (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      )
    )
    (1): BasicBlock(
      (conv1): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (relu): ReLU(inplace=True)
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    )
  )
  (7): Sequential(
    (0): BasicBlock(
      (conv1): Conv2d(256, 512, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), bias=False)
      (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (relu): ReLU(inplace=True)
      (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (downsample): Sequential(
        (0): Conv2d(256, 512, kernel_size=(1, 1), stride=(2, 2), bias=False)
        (1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      )
    )
    (1): BasicBlock(
      (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (relu): ReLU(inplace=True)
      (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    )
  )
  (8): AdaptiveAvgPool2d(output_size=(1, 1))
)
```

```
def get_embedding(img):
    with torch.no_grad():
        emb = model(img.unsqueeze(0))
    return emb.flatten().numpy()
```

```
import numpy as np

embeddings = []
images = []

for i in range(200):
    img, label = dataset[i]
    embeddings.append(get_embedding(img))
    images.append(img)

embeddings = np.array(embeddings)
```

```
from sklearn.metrics.pairwise import cosine_similarity
import matplotlib.pyplot as plt

sim = cosine_similarity(embeddings)

def show_similar(query_idx, top_k=5):
    similar_indices = sim[query_idx].argsort()[-top_k-1:-1][::-1]

    plt.figure(figsize=(12,3))

    # Query image
    plt.subplot(1, top_k+1, 1)
    plt.imshow(images[query_idx].permute(1,2,0))
    plt.title("Query")
```

```
    plt.axis('off')

    # Similar images
    for i, idx in enumerate(similar_indices):
        plt.subplot(1, top_k+1, i+2)
        plt.imshow(images[idx].permute(1,2,0))
        plt.title(f"Sim: {sim[query_idx][idx]:.2f}")
        plt.axis('off')

    plt.show()
```

```
show_similar(0)
```

```
print(sim[0].max(), sim[0].min())
```

```
0.99999976 0.6222115
```

```
!pip install git+https://github.com/openai/CLIP.git
```

```
Collecting git+https://github.com/openai/CLIP.git
  Cloning https://github.com/openai/CLIP.git to /tmp/pip-req-build-y8rg2kfd
  Running command git clone --filter=blob:none --quiet https://github.com/openai/CLIP.git /tmp/pip-req-build-y8rg2kfd
  Resolved https://github.com/openai/CLIP.git to commit d05afc436d78f1c48dc0dbf8e5980a9d471f35f6
  Preparing metadata (setup.py) ... done
Collecting ftfy (from clip==1.0)
  Downloading ftfy-6.3.1-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from clip==1.0) (26.0)
Requirement already satisfied: regex in /usr/local/lib/python3.12/dist-packages (from clip==1.0) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from clip==1.0) (4.67.3)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (from clip==1.0) (2.10.0+cu128)
Requirement already satisfied: torchvision in /usr/local/lib/python3.12/dist-packages (from clip==1.0) (0.25.0+cu128)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-packages (from ftfy->clip==1.0) (0.6.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch->clip==1.0) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch->clip==1.0) (4.15.0)
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Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch->clip==1.0) (12.8
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Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /usr/local/lib/python3.12/dist-packages (from torch->clip==1.0) (11.7
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in /usr/local/lib/python3.12/dist-packages (from torch->clip==1.0) (12.5
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch->clip==1.0) (0.7.1)
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Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings==12.9.4->torch->clip
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Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch->clip==1.0) (1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch->clip==1.0) (3.0.3)
Downloading ftfy-6.3.1-py3-none-any.whl (44 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 44.8/44.8 kB 3.9 MB/s eta 0:00:00
Building wheels for collected packages: clip
  Building wheel for clip (setup.py) ... done
  Created wheel for clip: filename=clip-1.0-py3-none-any.whl size=1369490 sha256=8414278d7dfc5b7d9c1c1c1e713226f8b1da0484ad6409fc93ba26
  Stored in directory: /tmp/pip-ephem-wheel-cache-qrrbiw20/wheels/35/3e/df/3d24cbfb3b6a06f17a2bfd7d1138900d4365d9028aa8f6e92f
Successfully built clip
Installing collected packages: ftfy, clip
Successfully installed clip-1.0 ftfy-6.3.1
```

```
import clip
import torch
```

```
device = "cuda" if torch.cuda.is_available() else "cpu"

model, preprocess = clip.load("ViT-B/32", device=device)
model.eval()
```

```
          (gelu): QuickGELU()
          (c_proj): Linear(in_features=2048, out_features=512, bias=True)
        )
        (ln_2): LayerNorm((512,), eps=1e-05, elementwise_affine=True)
      )
      (8): ResidualAttentionBlock(
        (attn): MultiheadAttention(
          (out_proj): NonDynamicallyQuantizableLinear(in_features=512, out_features=512, bias=True)
        )
        (ln_1): LayerNorm((512,), eps=1e-05, elementwise_affine=True)
        (mlp): Sequential(
          (c_fc): Linear(in_features=512, out_features=2048, bias=True)
          (gelu): QuickGELU()
          (c_proj): Linear(in_features=2048, out_features=512, bias=True)
        )
        (ln_2): LayerNorm((512,), eps=1e-05, elementwise_affine=True)
      )
      (9): ResidualAttentionBlock(
        (attn): MultiheadAttention(
          (out_proj): NonDynamicallyQuantizableLinear(in_features=512, out_features=512, bias=True)
        )
        (ln_1): LayerNorm((512,), eps=1e-05, elementwise_affine=True)
        (mlp): Sequential(
          (c_fc): Linear(in_features=512, out_features=2048, bias=True)
          (gelu): QuickGELU()
          (c_proj): Linear(in_features=2048, out_features=512, bias=True)
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        )
        (ln_1): LayerNorm((512,), eps=1e-05, elementwise_affine=True)
        (mlp): Sequential(
          (c_fc): Linear(in_features=512, out_features=2048, bias=True)
          (gelu): QuickGELU()
          (c_proj): Linear(in_features=2048, out_features=512, bias=True)
        )
        (ln_2): LayerNorm((512,), eps=1e-05, elementwise_affine=True)
      )
      (11): ResidualAttentionBlock(
        (attn): MultiheadAttention(
          (out_proj): NonDynamicallyQuantizableLinear(in_features=512, out_features=512, bias=True)
        )
        (ln_1): LayerNorm((512,), eps=1e-05, elementwise_affine=True)
        (mlp): Sequential(
          (c_fc): Linear(in_features=512, out_features=2048, bias=True)
          (gelu): QuickGELU()
          (c_proj): Linear(in_features=2048, out_features=512, bias=True)
        )
        (ln_2): LayerNorm((512,), eps=1e-05, elementwise_affine=True)
      )
    )
  )
  (token_embedding): Embedding(49408, 512)
  (ln_final): LayerNorm((512,), eps=1e-05, elementwise_affine=True)
)
```

```
import torchvision.datasets as datasets

dataset = datasets.CIFAR10(
    root='./data',
    download=True,
    transform=preprocess
)
```

```
import numpy as np

embeddings = []
images = []
labels = []

for i in range(200):  # keep small
    img, label = dataset[i]

    img_input = img.unsqueeze(0).to(device)

    with torch.no_grad():
        emb = model.encode_image(img_input)

    embeddings.append(emb.cpu().numpy().flatten())
    images.append(img)   # keep for visualization
```

```
    labels.append(label)

embeddings = np.array(embeddings)
```

```
from sklearn.metrics.pairwise import cosine_similarity

sim = cosine_similarity(embeddings)
```

```
import matplotlib.pyplot as plt

def show_similar(query_idx, top_k=5):
    similar_indices = sim[query_idx].argsort()[-top_k-1:-1][::-1]

    plt.figure(figsize=(12,3))

    # Query
    plt.subplot(1, top_k+1, 1)
    plt.imshow(images[query_idx].permute(1,2,0))
    plt.title("Query")
    plt.axis('off')

    # Results
    for i, idx in enumerate(similar_indices):
        plt.subplot(1, top_k+1, i+2)
        plt.imshow(images[idx].permute(1,2,0))
        plt.title(f"{sim[query_idx][idx]:.2f}")
        plt.axis('off')

    plt.show()
```

```
show_similar(0)
```

```
WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers).
WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers).
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WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers).
```

```
same_class = []
diff_class = []

for i in range(100):
    for j in range(i+1, 100):
        if labels[i] == labels[j]:
            same_class.append(sim[i][j])
        else:
            diff_class.append(sim[i][j])

print("Same class avg similarity:", sum(same_class)/len(same_class))
print("Different class avg similarity:", sum(diff_class)/len(diff_class))
```

```
Same class avg similarity: 0.8253408682227593
Different class avg similarity: 0.747512127086003
```

```
from sklearn.preprocessing import normalize

embeddings = normalize(embeddings)
sim = cosine_similarity(embeddings)
```

```
for i in [0, 5, 10]:
    show_similar(i)
```

```
WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers).
WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers).
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WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers).
```

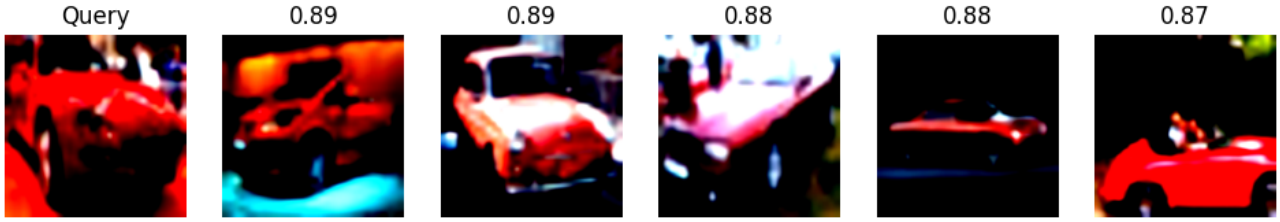


```
WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers).
WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers).
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WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers).
```

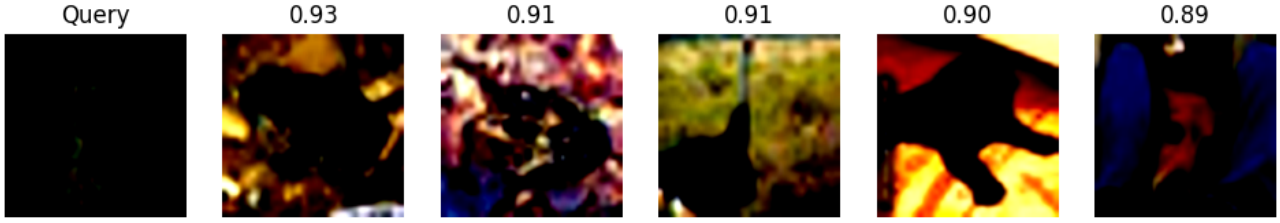

Start coding or generate with AI.